

Retail Orders Data Analysis using Python & SQL

Project Overview

This project focuses on analyzing retail order data using Python and SQL Server to generate meaningful business insights. The dataset was downloaded from Kaggle, cleaned and transformed using Python (Pandas), and then loaded into SQL Server for analytical querying.

The objective of this project is to demonstrate an end-to-end data analytics workflow including data extraction, data cleaning, feature engineering, database loading, and business reporting.

Business Problem

Retail businesses generate large volumes of sales data every day. Decision-makers need answers to questions such as:

- Which products generate the highest revenue?
- Which products perform best in different regions?
- How are sales growing year-over-year?
- Which categories contribute the most to revenue?
- Which sub-categories show the highest business growth?

This project aims to answer these business questions through data analysis.

Tools & Technologies Used

Python

- Pandas
- Kaggle API
- ZipFile
- SQLAlchemy

Database

- Microsoft SQL Server

Concepts Used

- Data Cleaning
 - Data Transformation
 - Feature Engineering
 - SQL Aggregations
 - Common Table Expressions (CTEs)
 - Window Functions
 - Business Analytics
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Data Processing Steps

1. Data Extraction

- Downloaded dataset from Kaggle using Kaggle API.
- Extracted CSV file from ZIP archive.

2. Data Cleaning

- Replaced invalid values such as:
 - Not Available
 - unknown
- Standardized column names by:
 - Converting to lowercase
 - Replacing spaces with underscores

3. Feature Engineering

Created new business metrics:

- Discount Amount
- Sale Price
- Profit

Formula Used:

Profit = Sale Price – Cost Price

4. Data Transformation

- Converted order_date into datetime format.
- Removed unnecessary columns after calculations.

5. Database Loading

- Connected Python with SQL Server using SQLAlchemy.

- Loaded transformed data into SQL Server table.
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Business Questions Solved

1. Top 10 Revenue Generating Products

Identified products contributing the highest sales revenue.

Business Value:

- Helps prioritize high-performing products.
 - Supports inventory planning and marketing strategies.
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2. Top 5 Products by Region

Found the best-selling products within each region using SQL Window Functions.

Business Value:

- Understand regional customer preferences.
 - Optimize regional inventory allocation.
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3. Month-over-Month Sales Comparison (2022 vs 2023)

Compared monthly sales performance across years.

Business Value:

- Track business growth trends.
 - Identify seasonal demand patterns.
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4. Highest Sales Month for Each Category

Determined the month in which each product category achieved maximum sales.

Business Value:

- Improve promotional planning.
 - Forecast future category performance.
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5. Highest Growth Sub-Category (2023 vs 2022)

Calculated year-over-year growth and identified the fastest-growing sub-category.

Business Value:

- Discover emerging product segments.
 - Support strategic investment decisions.
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Key SQL Concepts Demonstrated

- GROUP BY
 - Aggregate Functions
 - CASE Statements
 - Common Table Expressions (CTE)
 - ROW_NUMBER()
 - Window Functions
 - Date Functions
 - Ranking Techniques
 - Year-over-Year Analysis
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Learning Outcomes

Through this project, I learned:

- End-to-end data analytics workflow
 - Data cleaning and preprocessing using Python
 - SQL Server integration with Python
 - Writing analytical SQL queries
 - Solving real-world business problems using data
 - Generating actionable business insights
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Conclusion

This project demonstrates how raw retail sales data can be transformed into valuable business insights using Python and SQL. The analysis helps stakeholders understand product performance, regional trends, revenue growth, and category-wise opportunities, enabling data-driven decision-making.